

Alexandra Verena Rottenkolber

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Personal Website — GitHub — LinkedIn

EDUCATION

Institute for Analytical Sociology, Linköping University, Sweden 2022 – 2026 (*expected*)
PhD in Analytical Sociology (Computational Social Science)
Focus: Computational modeling of social mechanisms, network inference, and large-scale text analysis
Supervisors: Károly Takács, Martin Arvidsson

Cornell University, USA Aug 2025 – Dec 2025
Visiting PhD Researcher at the Department of Sociology & Information Science
Hosted by Michael Macy and the Social Dynamics Lab

University of Oxford, UK 2020 – 2021
MSc in Social Data Science, *Distinction*
Courses in NLP, Applied ML, Network Analysis, Statistics & Causal Inference

Hertie School of Governance, Germany 2019 – 2022
MPP in Public Policy (Quantitative Policy Analysis), *Grade: 1.4* (“very good”; top grade band)

University of Heidelberg, Germany 2014 – 2018
BSc in Physics and Geography, *Grade: 1.5* (“very good”; top grade band)

RESEARCH EXPERIENCE

Max Planck Institute for Demographic Research, Germany Jun 2024 – Aug 2024
Social Data Science Researcher

- Lead author of a cross-institutional project on network and gender dynamics in migration, coordinating 7 researchers
- Built scalable R/Python pipelines analysing 20M+ articles and 8M+ authors
- Results shown at three international conferences (IC2S2, PAA, INAS, 2025)

Michigan State University, USA & University of Oxford, UK Jun 2022 – Jul 2023
Graduate Research Associate – Quello Center for Media & Information Policy

- Investigated digital inequalities using web-scraping and network distance measures
- Applied graph algorithms to compare network topologies over time

University of Oxford, Dept. of Social Policy & Intervention, UK Aug 2021 – Feb 2022
Graduate Research Assistant – ERC Project “Changing Elites”

- Developed OCR and NLP workflows linking 16M probate records with 66M death registrations across 130 years, creation of a novel scientific dataset

TEACHING & MENTORSHIP

Master Thesis Supervision, Linköping University, Sweden 2023, 2024, 2025
Supervised 3 MSc CSS theses in computational text analysis

SICSS Summer Institute in Computational Social Science, Linköping University, Sweden 2023 & 2025
Assistance for Workshop on Social Network Analysis

University of Oxford, Oxford Internet Institute, UK 2023 – 2024
Teaching Assistant, Social Network Analysis (Graduate level), developed teaching materials & assisted course design.

Swedish Interdisciplinary Graduate Research School in CSS, Linköping University, Sweden 2023
Assistant for Social Network Analysis & Machine Learning (Post-Graduate level)

INDUSTRY & POLICY EXPERIENCE

German Corporation for International Cooperation (GIZ) GmbH, Germany May 2020 – Sep 2020
Independent Consultant, Climate Policy
Supported climate governance projects; managed multi-stakeholder coordination.

Contributed to the development of crisis forecasting dashboard using Python and RShiny.

SERVICE & ENGAGEMENT

Register Data Contact Person – Institute for Analytical Sociology, Linköping University Nov 2024 – today
Contact person for the Swedish national population register data, ensuring data compliance and reproducibility.

Workshop Organizer – Swedish Language Technology Conference (SLTC) 2024 2024
Lead organiser of a Computational Social Science & Language Technology workshop with 40+ participants.

PhD Student Representative – Institute for Analytical Sociology, Linköping University Oct 2022 – Oct 2024
Serving as a PhD representative

Data4Good Volunteer – CorrelAid x Audiopedia 2021
Assisted developing a data pipeline for mental health information in audio format for illiterate people.

Organisation for student exchange in Middle and Eastern Europe GFPS Aug 2013 – Sep 2020
(Gemeinschaft für studentischen Austausch in Mittel- und Osteuropa, GFPS e.V.)
Supervision of scholarship holders of the GFPS e.V., development of the Cooperate Design, organisation of a language course and intercultural seminars

AWARDS & SCHOLARSHIPS

- Linköping University Travel Grant – Research stay at Cornell University
- German Academic Scholarship Foundation (Studienstiftung des Deutschen Volkes) – *awarded for academic excellence to fewer than 0.5% of German students*
- DAAD Scholarship for a master's degree programme abroad
- DAAD Go East Scholarship

SELECTED RESEARCH PROJECTS & WORKING PAPERS

The Shielding Effect: How workplace segregation can countervail occupational segregation
Alexandra Rottenkolber, Martin Arvidsson, Maria Brandén, and Károly Takács – *Under Review & Resubmit (ESR)*.
Uses the Swedish population-scale register data to test mechanisms of gendered labour sorting. Available upon request.

The role of professional networks and institutional prestige in shaping the first career moves of scholars
Alexandra Rottenkolber, Ola Ali, Gergely Mónus, Jiaxuan Li, Jisu Kim, Daniela Perrotta, and Aliakbar Akbaritabar – *Submitted to PNAS Nexus, preprint available here*. Study on how a scholars' personal and institutional embedding shapes their career mobility.

Do employers practice what they preach? Parenthood bias among employers in hiring, self-presentation, and self-perception in Germany
Alexandra Rottenkolber* & Isabel Habicht* (*equal contributions) – *in progress*. A correspondence experiment, paired with job-ad text analysis and an HR survey, to test whether employers' signalling and rhetoric diverge from callbacks.

LANGUAGES & TECHNICAL SKILLS

Techniques: Web-scraping, Statistical Analysis, NLP, Network Analysis, Machine Learning, API integration, Visualisation

Programming Languages: Python, R, SQL

Libraries: pandas, polars, networkx, tidyverse, igraph, matplotlib, ggplot, gensim, familiarity with scikit-learn, TensorFlow and PyTorch

Tools & Collaboration: GitHub, Markdown, L^AT_EX, QGIS, ArcGIS, Adobe Illustrator

Spoken Languages: German (native), English (fluent)

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